



2004
HSC COURSE
ASSESSMENT TASK 3

Mathematics

General Instructions

- Time allowed – 50 minutes
- Write using blue or black pen
- Board-approved calculators and mathaids may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question
- Each question is to be answered on a separate answer page.

Topics:

- Trigonometric Functions
- Exponential and Logarithmic Functions

Total Marks – 34

- Attempt Questions 1 – 4
- Question 1 – 8 marks
Question 2 – 9 marks
Question 3 – 8 marks
Question 4 – 9 marks

Question 1 (8 marks) Use a SEPARATE answer page.

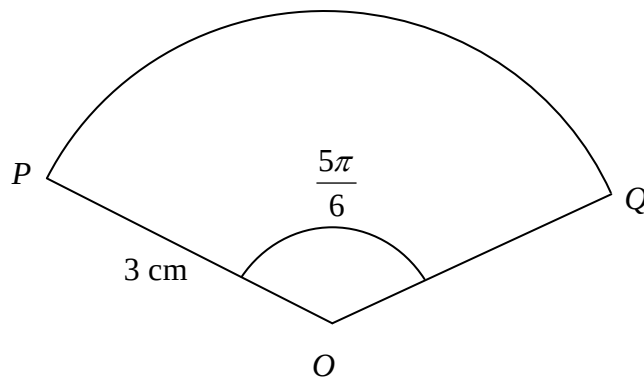
Marks

(a) Write down the exact value of 120° in radians. **1**

(b) Find the exact value of $\cos \frac{\pi}{4}$. **1**

(c) Solve $2 \cos x + \sqrt{3} = 0$ for $0 \leq x \leq 2\pi$. **2**

(d)



In the diagram, PQ is an arc of a circle with centre O . The radius $OP = 3$ cm and the angle POQ is $\frac{5\pi}{6}$ radians. Find the exact area of the sector POQ . **2**

(e) Evaluate $\int_0^{\frac{\pi}{6}} 2 \sin 2x \, dx$ **2**

Question 2 (9 marks) Use a SEPARATE answer page. **Marks**

(a) Differentiate $\frac{\sin x}{x}$. **2**

(b) Find $\int \sec^2(1+3x) dx$. **1**

(c) Find the equation of the tangent to the curve $y = 1 - 2 \cos x$ at the point
where $x = \frac{\pi}{2}$. **3**

(d) (i) Sketch $y = 2 \cos \frac{x}{2}$, for $0 \leq x \leq 2\pi$, showing all essential features and
labelling the curve. **2**

(ii) Hence clearly sketch $y = 1 + 2 \cos \frac{x}{2}$ on the same set of axes, labelling
this curve also. **1**

Question 3 (8 marks) Use a SEPARATE answer page. **Marks**

(a) Evaluate, correct to three significant figures:

(i) $e^{-1.9}$ **1**

(ii) $\log_e 36$. **1**

(b) Solve $2\log_5 3 = \log_5 x - \log_5 9$. **2**

(c) Find $\int \frac{1}{5-2x} dx$. **1**

(d) Differentiate $3e^{x^2}$. **1**

(e) (i) Show that the point $(e+2, 1)$ lies on the curve $y = \log_e(x-2)$. **1**

(ii) Sketch the graph of $y = \log_e(x-2)$, showing clearly any asymptotes. **1**

Question 4 (9 marks) Use a SEPARATE answer page.

Marks

- (a) A solid is formed by rotating the portion of the curve $y = e^x$ between $x = 0$ and $x = 2$ around the x -axis.

3

Find the volume of this solid. Leave the answer in exact form.

(b)

- (i) Show that $\frac{d}{dx}(xe^{-x}) = (1-x)e^{-x}$.

1

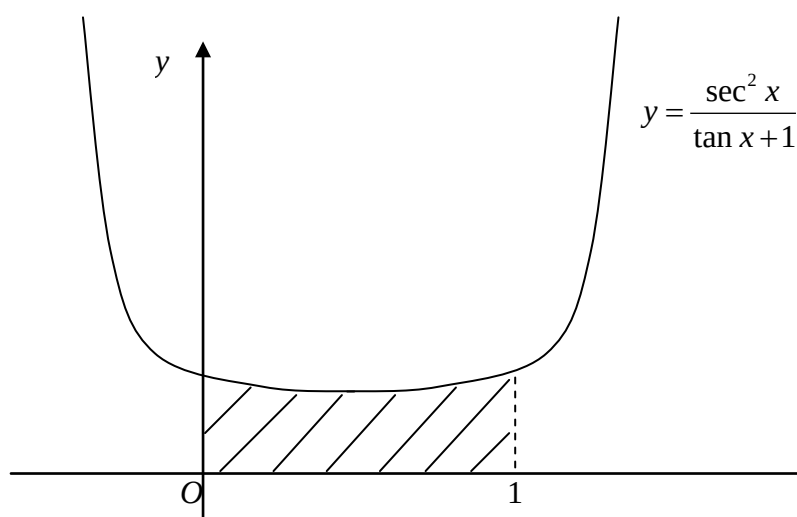
- (ii) Find the stationary point on the curve $y = xe^{-x}$.

1

- (iii) Determine the nature of this stationary point.

1

(c)



The shaded region in the diagram is bounded by the curve $y = \frac{\sec^2 x}{\tan x + 1}$, the x -axis, the y -axis, and the line $x = 1$. Find the area of the shaded region.

3

End of Paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

HSC MATHS 2004 - T3

Q1

(a) $120^\circ = \frac{2\pi}{3}$

1

(b) $\cos \frac{\pi}{4} = \frac{1}{\sqrt{2}}$

1

(c) $2 \cos x + \sqrt{3} = 0 \quad 0 \leq x \leq 2\pi$

$\cos x = -\frac{\sqrt{3}}{2}$
 $x = \frac{5\pi}{6}, \frac{7\pi}{6}$



1 for one solution
 2 - both "s

(d) Area = $\frac{1}{2} r^2 \theta$
 $= \frac{1}{2} \times 3^2 \times \frac{5\pi}{6}$
 $= \frac{15\pi}{4} \text{ cm}^2$

1 for correct substitution.

1 for correct answer with units.

(e) $\int_0^{\pi/6} 2 \sin 2x \, dx = \left[-2 \cos 2x \times \frac{1}{2} \right]_0^{\pi/6}$

$= [-\cos 2x]_0^{\pi/6}$

1 for correct primitive

$= -\cos \frac{\pi}{3} - (-\cos 0)$

1 for correct answer.

$= -\frac{1}{2} + 1$

$= \frac{1}{2}$

Q2.

(a) $\frac{d}{dx} \left(\frac{u}{v} \right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$

2 for correct answer

$\frac{d}{dx} \left(\frac{\sin x}{x} \right) = \frac{x \cos x - \sin x \times 1}{x^2}$

$= \frac{x \cos x - \sin x}{x^2}$

1 for correct application of rule except for 1 error?

Q2 cont'd.

(b) $\int \sec^2(1-3x) dx = -\frac{1}{3} \tan(1-3x) + C$

1

(c) $y = 1 - 2 \cos x$
 $y' = 2 \sin x$

at $x = \frac{\pi}{2}$, $y = 1 - \cos \frac{\pi}{2} = 1$

$y' = 2 \sin \frac{\pi}{2} = 2 = m$

$y - y_1 = m(x - x_1)$

$y - 1 = 2(x - \frac{\pi}{2})$

$y = 2x + \pi + 1$

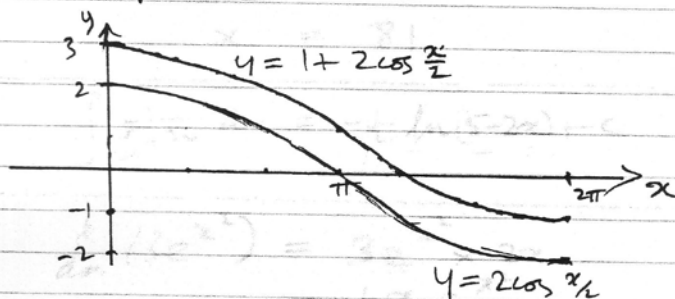
1 for correct gradient

1 for correct sub'n into formula.

1 for correct answer

(d) $y = 2 \cos \frac{x}{2}$ $0 \leq x \leq 2\pi$

(i) Period = $\frac{2\pi}{1/2} = 4\pi$
 Amplitude = 2



2 for correctly drawn & labelled graph

1 for graph showing either correct amplitude or correct period.

(ii) See graph.

1 for correct graph, based on result of (i)

Q3.

(a) (i) $e^{-1.9} = 0.150$ (3.s.f.)

1

(ii) $\log_e 36 = 3.58$ (3.s.f.)

1

(b) $2 \log_5 3 = \log_5 x - \log_5 9$

$\log_5 3^2 = \log_5 \frac{x}{9}$

$9 = \frac{x}{9}$

$x = 81$

2 for correct answer

1 for correct application of at least 1 log law.

(c) $\int \frac{1}{5-2x} dx = -\frac{1}{2} \ln(5-2x) + C$

1

(d) $\frac{d}{dx} (3e^{x^2}) = 3e^{x^2} \times 2x$
 $= 6xe^{x^2}$

1

(e) (i) $y = \log_e (x-2)$ sub in $(e+2, 1)$

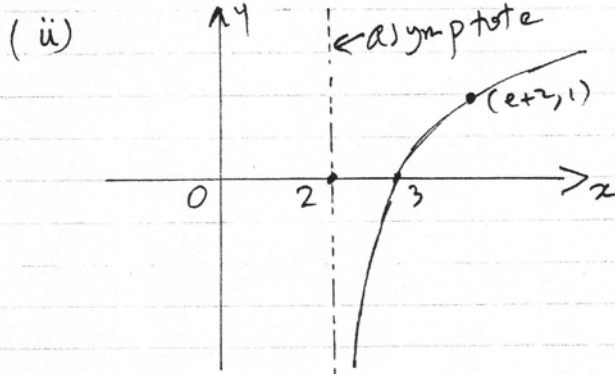
$= \log_e (e+2-2)$

$= \log_e e$

$= 1$

$\therefore (e+2, 1)$ lies on $y = \log_e (x-2)$

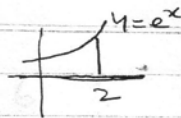
1



1 for showing correctly shaped curve & clearly indicating asymptote

Q4.

(a) $V = \pi \int_a^b y^2 dx$



$$= \pi \int_0^2 (e^x)^2 dx$$

$$= \pi \int_0^2 e^{2x} dx$$

$$= \frac{\pi}{2} [e^{2x}]_0^2$$

$$= \frac{\pi}{2} (e^4 - e^0) = \frac{\pi}{2} (e^4 - 1)$$

1 for correct
volume
equation

1 for correct
primitive

1 for correct
evaluation

(b) (i) $y = xe^{-x}$ $uv' + vu'$
 $y' = x \times (-e^{-x}) + e^{-x} \times 1$
 $= e^{-x} (1 - x)$

(ii) let $e^{-x} (1 - x) = 0$
 $x = 1$ ($e^{-x} \neq 0$)
 $y = 1 \times e^{-1}$
 $= 1/e$

$\therefore$ Stationary point $= (1, 1/e)$

(iii) $x < 1$ $1 > 0$
 $e^{-x} (1 - x)$ $(+)$ 0 $(-)$

$\therefore (1, 1/e)$ is MAXIMUM turning point

Must show
working

(c) $A = \int_0^1 \frac{\sec^2 x}{\tan^2 x + 1} dx$

$$= [\ln(1 + \tan x)]_0^1$$

$$= (\ln(1 + \tan 1)) - \ln(1 + \tan 0)$$

$$= \ln(1 + \tan 1) - 0$$

$$= 0.9389 \dots$$

$$= 0.94 \text{ (2 dec. pl.)}$$

1 for correct
integral
equation

1 for correct
primitive

1 for correct
evaluation